

Hollow Enemy AI Tooling Recommendations

Decision memo - snapshot May 2, 2026 - optimized for long-term power with minimal per-enemy setup

Decision

Buy or adopt pathfinding before buying a behavior authoring suite. Hollow already has a good data spine through EnemyDefinition, EnemyBehaviorId, intelligence, disposition, attack budgets, and room-local collision. The missing production multiplier is route planning around authored room obstacles, holes, and future perception spaces.

- Best paid first purchase: A* Pathfinding Project Pro.
- Best no-new-purchase first pass: Unity AI Navigation, already installed locally.
- Best visual behavior-tree layer later: Behavior Designer Pro for long-term power, or NodeCanvas if we prefer mature GameObject-first authoring and full source.
- Avoid Emerald AI as the core Hollow enemy system; it is better for generic NPC projects than bespoke room combat.

Current Hollow Baseline

- Unity 6000.4.1f1 with com.unity.ai.navigation 2.0.11 already in Packages/manifest.json.
- Enemies currently use direct vector chase/retreat/hold in EnemyRuntimeController and RoomLocalCollision.
- EnemyDefinition already carries archetype, behavior, movement mode, intelligence, disposition, range, attack, charge, body, and split data.
- Existing M72 docs explicitly defer pathfinding, line of sight, squad coordination, and richer boss behavior.

Recommendation Shape

Phase	Action	Why
1 - Navigation	Use A* Pro if buying; otherwise prototype with Unity AI Navigation.	Solves obstacle-aware movement first and keeps combat behavior stable.
2 - Behavior abstraction	Extract movement decisions into data-driven movement intents and add perception hooks.	Keeps tests deterministic while preparing for richer AI.
3 - Visual authoring	Add Behavior Designer Pro or NodeCanvas only if shared graphs will accelerate enemy/boss design.	Avoids per-enemy graph work while allowing designer-facing logic.

Ranked Tools

1. A* Pathfinding Project Pro

Snapshot	Asset Store fallback: v5.4.6, Jan 22 2026, \$140, original Unity 2021.3.45.
Why	Best long-term pathfinding upgrade: grid/recast graphs, graph updates, AIPath-style movement, and Pro local avoidance using ORCA/RVO.
Hollow fit	Excellent. Solves Hollow's real missing layer: obstacle-aware paths, room graph reuse, dynamic updates, and local avoidance without replacing combat data.
Codex/setup	User purchases/imports. Codex can build RoomNavigationBuilder, an enemy navigation adapter, tests, and one shared prefab/spawn integration.
Risks	Paid dependency and import/API validation. Wrap behind a Hollow interface so Unity AI Navigation/current movement can remain fallback.
Verdict	Best paid pick. If we buy one tool, buy this first.

2. Unity AI Navigation

Snapshot	Already installed locally as com.unity.ai.navigation 2.0.11.
Why	No new purchase, official Unity support, runtime/edit-time NavMesh building, dynamic obstacles, and links.
Hollow fit	Good baseline for X/Z room spaces with generated floor and obstacle geometry.
Codex/setup	Codex can implement runtime NavMesh building, an agent adapter, and shared enemy movement without per-enemy setup.
Risks	Less grid-native than Hollow room data. Runtime surface filters and generated-room baking need care.
Verdict	Best no-cost prototype if we are unsure about buying A* Pro.

3. Behavior Designer Pro

Snapshot	Asset Store fallback: v2.1.12, Jan 28 2026, \$145, original Unity 2022.3.20.
Why	Strong visual behavior-tree authoring with DOTS-backed traversal, shared variables, subtrees, debugging, and custom tasks.
Hollow fit	Strong after pathfinding for readable sentinel/prey/predator/charger/turret/boss decisions.
Codex/setup	User purchases/imports. Codex can create reusable GameObject tasks bound to EnemyDefinition and Hollow attack budgets.
Risks	Adds Entities/Burst dependency and Opsive documents no WebGL support while Entities/Burst lack it. Does not solve navigation alone.
Verdict	Best visual behavior-tree option for long-term power, but phase 3 rather than phase 1.

4. NodeCanvas

Snapshot	Marketplace/fallback: v3.4.1, Feb 20 2026, list \$120 with sale pricing sometimes visible.
Why	Mature BT/FSM/Dialog authoring, full source, blackboards, subgraphs, runtime debugging, and A* integration signal.
Hollow fit	Very good if we want designer-facing logic without DOTS assumptions; especially attractive for bosses and hybrid state behavior.
Codex/setup	User purchases/imports. Codex can write reusable tasks and map variables from EnemyDefinition.
Risks	Less future-looking for massive agent counts. Graph freedom can invite one-off per-enemy authoring unless constrained.
Verdict	Best practical alternative to Behavior Designer Pro, especially if DOTS/WebGL risk matters.

5. Unity Behavior

Snapshot	Unity docs: com.unity.behavior 1.0.13 released for Unity Editor 6000.0.
Why	Official graph-based behavior trees with reusable subgraphs, C# integration, prebuilt nodes, and play-mode debugging.
Hollow fit	Interesting no-purchase behavior layer after navigation is solved.
Codex/setup	Codex can add the package if requested, create custom nodes, and bind shared graphs to current enemy data.
Risks	Younger ecosystem than Behavior Designer Pro or NodeCanvas; likely needs more Hollow-specific nodes.
Verdict	Good official experiment. Not the top production recommendation yet.

6. GOAP v3 by CrashKonijn

Snapshot	Asset Store fallback: free, v3.1.1, Dec 12 2025; docs show package tag 3.1.2.
Why	Powerful multi-threaded goal/action planning with debugging for systemic AI.
Hollow fit	Situational. Useful for future bosses, companions, or NPC planners, not first for obstacle-aware chasers.
Codex/setup	Codex can integrate and author code-based goals/actions, but the goal model is design-heavy.
Risks	Adds planning complexity where Hollow mostly needs movement and readable tactical variants.
Verdict	Keep in reserve for systemic AI.

7. Utility Intelligence GO v3

Snapshot	Asset Store fallback: v3.1.3, Apr 18 2026, \$125, original Unity 6000.0.71.
Why	Utility scoring fits target choice, hold range, retreat, pressure, charge use, and boss phase selection.
Hollow fit	Useful later as a high-level decision layer for Tactical/Cunning enemies; not navigation.
Codex/setup	User purchases/imports. Codex can bind scoring inputs to EnemyDefinition, room state, distance, damage memory, and attack budget.
Risks	Newer/smaller market signal. Scoring curves need playtest tuning.
Verdict	Interesting later. Do not buy before pathfinding.

8. Emerald AI 2025

Snapshot	Marketplace: v1.3.3, Apr 8 2026, list \$60 with \$30 sale snapshot, Unity 6000.0.21f1 compatible.
Why	Fast generic RPG/NPC/animal/faction/patrol/combat setup.
Hollow fit	Poor as Hollow's core layer because it overlaps with bespoke room combat, readability, budgets, bosses, spawning, and definitions.
Codex/setup	User purchases/imports. Codex could adapter-wrap it, but much of the value is inspector-driven per-enemy setup.
Risks	Duplicate combat logic, integration overhead, and likely manual tuning per enemy.
Verdict	Do not use as the core Hollow enemy AI solution.

Manual Setup Answer

Codex can take care of most integration work once assets exist in the project: shared components, adapters, custom graph tasks/nodes, prefab/spawn glue, ScriptableObject bindings, migration helpers, and tests.

The user still needs to purchase/import paid Asset Store packages. For visual graph tools, the user may also want to inspect or tune shared behavior graphs in Unity. The recommended architecture avoids per-enemy manual configuration by using existing EnemyDefinition data and shared prefabs/subgraphs.

Sources

Source	URL
A* Pathfinding Project Pro Asset Store	https://assetstore-fallback.unity.com/packages/tools/behavior-ai/a-pathfinding-project-pro-87744
A* graph types	https://arongranberg.com/astar/documentation/stable/graphtypes.html
A* local avoidance	https://arongranberg.com/astar/documentation/stable/localavoidance.html
Unity AI Navigation manual	https://docs.unity3d.com/ja/current/Manual/com.unity.ai.navigation.html
Unity Behavior manual	https://docs.unity3d.com/kr/current/Manual/com.unity.behavior.html
Behavior Designer Pro Asset Store	https://assetstore-fallback.unity.com/packages/tools/visual-scripting/behavior-designer-pro-dots-powered-behavior-trees-298743
Behavior Designer Pro docs	https://opsive.com/support/documentation/behavior-designer-pro/
Behavior Designer Pro requirements	https://opsive.com/support/documentation/behavior-designer-pro/requirements/
NodeCanvas Asset Store	https://marketplace.unity.com/packages/tools/visual-scripting/nodecanvas-14914
NodeCanvas official site	https://nodecanvas.paradoxnotion.com/
GOAP v3 Asset Store	https://assetstore-fallback.unity.com/packages/tools/behavior-ai/goap-v3-302434
GOAP v3 docs	https://goap.crashkonijn.com/readme/tutorial/gettingstarted
Utility Intelligence GO v3 Asset Store	https://assetstore-fallback.unity.com/packages/tools/behavior-ai/utility-intelligence-go-v3-utility-ai-framework-308338
Emerald AI 2025 Asset Store	https://marketplace.unity.com/packages/tools/behavior-ai/emerald-ai-2025-268519

Asset Store prices and versions are checkout snapshots from May 2, 2026 and should be verified before purchase.